Document 3: Vector Databases and Similarity Search

1. Purpose
Vector databases are engineered to store, index, and query high-dimensional vector embeddings efficiently. Unlike relational databases, they look for geometric proximity rather than exact keyword matches.

2. Popular Distance Metrics
- Cosine Similarity: Measures the angle between two vectors, invariant to magnitude.
- Euclidean Distance: Computes the straight-line distance between two points in Euclidean space.
- Dot Product: Measures magnitude and direction alignment.